Toulouse School of Economics, 2011-2012
Macroeconomics I – Franck Portier & Gilles Saint Paul

Exam

This is a 3 hours exam. All documents allowed.

I – True, False, Uncertain? (*1/3 of the points*)

For each of those statements, say whether it is true, false, or uncertain and explain why. You should target around 10 lines per question.

1 – In an OLG model, no growth trajectory Pareto-dominates the Golden Rule steady state.

2 – As long as unemployment remains positive, it is efficient to reduce it.

3 – In the Mortensen-Pissarides model, the stock of vacancies is a state variable.

4 – The long-term interest rate in the Ramsey model is pinned down by the Euler equation for consumption.

5 – Neo-classical growth models predict rapid convergence in GDP per capita between countries.

6 – The augmented Solow model predicts unconditional converge.

7 – In endogenous growth models, better enforcement of intellectual property rights unambiguously raises welfare.

8 – An unfunded pension system reduces capital accumulation.

9 – A funded pension system investment in government debt is subject to Ricardian equivalence and has no effect on the allocation of resources.

10 – According to the Hosios conditions, the labor income share in total GDP should equal the elasticity of unemloyment in the matching function.

II – OLG Model with Human Capital (*1/3 of the points*)

People live two periods. Each period there is a cohort of size N being born. When young, people allocate their time between work and investment in the human capital of their children. They consume and save. With their savings they accumulate physical capital. As in the Diamond OLG model, the physical capital accumulated by the young at t enters the production function of at $t+1$ and has a $t+1$ time index. When old, they do not work and just consume. The labor endowment of an individual born at t is given by h_t, his human capital level. The wage of human capital is denoted ω_t. The total return on capital between t and $t+1$ is $1+r_{t+1}$. The fraction of time spent educating children by generation t is denoted by x_t: The human capital of a child of generation $t+1$ is

$$h_{t+1} = Bh_t x_t,$$

where B is a constant. The production function for the final good is

$$Y_t = AK_t^{\alpha} L_t^{1-\alpha},$$

where K_t is the capital stock and $L_t = H_t(1-x_t) = Nh_t(1-x_t)$ is the total labor input. There is full depreciation of capital between two periods. The utility of a member of the cohort born at t is given by

$$U = \ln c_{yt} + \gamma \ln h_{t+1} + \phi \ln c_{ot+1},$$

where c_y is consumption when young and c_o is consumption when old, and h_{t+1} is offspring human capital.

1 – Compute the return to human and physical capital at t, as a function of K_t, H_t and x_t.

2 – Show that the optimal savings by generation t is given by

$$s_t = \frac{\phi}{1+\phi+\gamma}\omega_t h_t$$

while the optimal value of x_t is

$$x_t = \frac{\gamma}{1+\phi+\gamma}$$

3 – Show that in the aggregate, the economy behaves as follows:

$$\begin{aligned} K_{t+1} &= QK_t^{\alpha}H_t^{1-\alpha} \\ H_{t+1} &= RH_t \end{aligned}$$

where Q and R are constants which you will express as functions of the model's parameters.

4 – What is the long-run growth rate of output in this economy?

5 – Characterize the transitional dynamics of the K/H ratio.

Assume now that there is a public pension system which taxes labor income at a constant rate τ and pays a pension p_t to each person of the old generation at t.

6 – Compute the value of s_t and x_t as a function of ω_t, h_t, r_{t+1} and p_{t+1}.

7 – What is the value of x in a balanced growth path? How is it affected by τ ?

8 – What is the effect of τ on long-run growth? Explain.

9 – What is the tax rate which delivers the maximum possible growth rate? Is it optimal from a welfare point of view?

10 – Derive an equation allowing to compute the long-term capital/human capital ratio. Assuming $\tau << 1$, what is the effect of the tax rate on that ratio? Explain.

III – RBC Model (*1/3 of the points*)

The following excerpts are taken from an article of Timothy Cogley and James M. Nason entitles "Output Dynamics in Real-Business-Cycle Models" and published in The American Economic Review, Vol. 85, No. 3 (Jun., 1995), pp. 492-511.

1 – The left panel of Figure 1 reports the autocorrelation function (ACF) for U.S. real per capita GNP growth, 1954:1-1988:4, along with bands that mark plus and minus two standard errors. The right panel of Figure 1 shows the spectrum for output growth, which was estimated by smoothing the periodogram using a Bartlett window. Present the results displayed on Figure 1. Why are those statistical moments meaningful?
- The left panel shows that output growth is serially correlated: if growth is high, it is likely to be also high the next quarter. There is serial correlation at order 1 and two, not after.
- The right panel displays an estimate of the spectral density of output growth, which indicates the part of the variability of output growth attributable to each frequency (or equivalently each period). A peak in the spectrum indicates that the corresponding periodic components have greater amplitude than other components and therefore contribute a greater portion of the variance. We do observe the typical shape of economic time series: the spectrum for output growth has a broad peak that ranges from approximately 2.33 to 7 years per cycle, with maximum power at roughly 3.2 years per cycle. Thus a relatively large proportion of the variance of output growth occurs at business-cycle frequencies.
- Those statistical moments are meaningful in that they show that the business cycle is (a) an important dimension of economic time series, (b) possesses some structure to be modeled (not a white noise)

2 – How do economists define and measure the Business Cycle?

- Let us consider a times series x_t that we want to decompose into a trend x_t^T and a cycle x_t^C.
- The business cycle is vaguely defined as the periodic but irregular up-and-down movements in economic activity, measured by fluctuations in real GDP and other macroeconomic variables.
- Statistical theory tells us that there is a infinity of decomposition of x_t into a stationary (typically "the cycle") and a non-stationary ("the trend").
- Expert analysis (NBER, business analysts, etc...) consider that there is a 2 to 8 years cycle in economic activity.
- One can then somewhat arbitrarily decide that the business cycle will be measured as the sum of all the periodic components of period less than 32 quarters (Hodrick-Prescott filter), or that components between 8 and 32 quarters only will be kept (band pass filter).
- There are several other ways of extracting a business cycles, some more structural (output gap measures), some not (growth rates, linear detrending).
- Generically, the business cycle is one filtered version of x_t.

3 – Extract 1 presents the Christiano-Eichenbaum model (Lawrence J. Christiano and Martin Eichenbaum, "Current Real-Business-Cycle Theories and Aggregate Labor-Market Fluctuations", The American Economic Review, June 1992, 82(3), pp. 430-50). Derive the fist-order conditions of the Social Planner problem in this economy. Discuss those conditions.

The Social Planner problem writes

$$\max_{c,n,k} E_t \left\{ \begin{array}{l} \sum_{j=0}^{\infty} \beta^j \left(\ln c_{t+j} + \gamma(N - n_{t+j})\right) \\ + \sum_{j=0}^{\infty} \beta^j \lambda_{t+j} \left(k_{t+j}^{\theta}(a_{t+j} n_{t+j})^{1-\theta} - c_{t+j} - k_{t+j+1} + (1-\delta)k_{t+j} - g_{t+j}\right) \end{array} \right\}$$

where λ is the undiscounted Lagrange multiplier of the resource constraint.
First order conditions write for period t (eliminating λ)

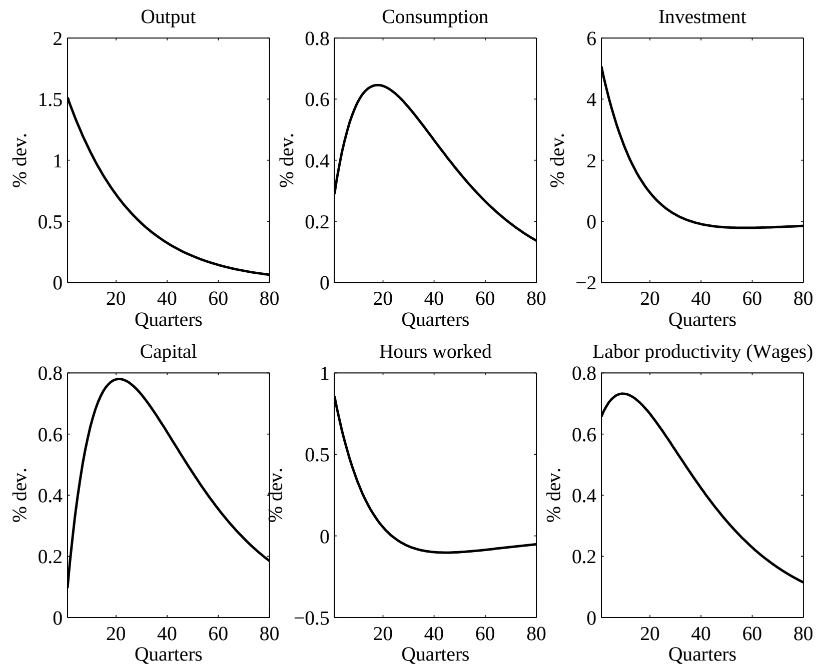
$$\begin{aligned} \gamma &= \frac{1}{c_t} \times (1-\theta) k_t^{\theta} a_t^{1-\theta} n_t)^{-\theta} \\ \frac{1}{c_t} &= \beta E_t \frac{1}{c_{t+1}} \left(\theta k_{t+1}^{\theta-1} (a_{t+1} n_{t+1})^{1-\theta} + 1 - \delta\right) \end{aligned}$$

First equation shows that marginal disutility of labor equals marginal productivity of labor times marginal utility of consumption. Second equation equalizes in expectations marginal utility of consumption today with discounted marginal utility of consumption tomorrow times the marginal return of saving one extra unit of capital today. Note that g does not affect marginal conditions. These allocations can be attained in competitive equilibrium if government spending are financed by lump-sum taxes.

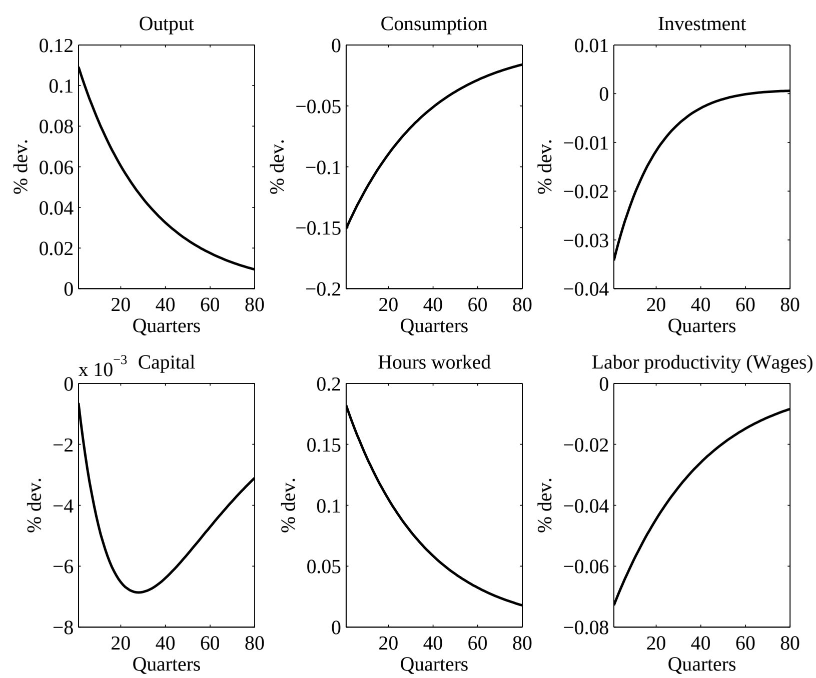
4 – What are the typical responses of consumption, output, investment and hours worked to respectively ε_a and ε_g shocks?

- Typical response to an autocorrelated technology shock: labor becomes more productive at the margin – labor demand shifts upwards – substitution effect dominates income effect so that labor supply increases –

consumption smoothing implies that consumption *and* investment increases.



- Typical response to an autocorrelated government spending shock : government takes resources – agents are poorer – negative wealth effect: consumption (today), consumption tomorrow (investment) and leisure decrease, so that output increases – there is a crowding out of private spending (consumption and investment) by public ones.



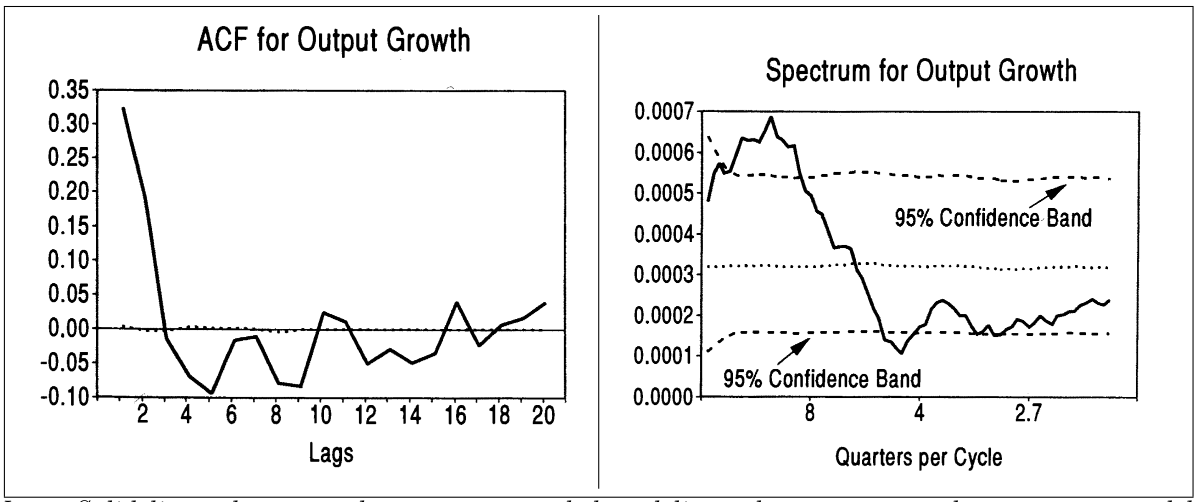
5 – The dotted lines in Figure 1 are estimated from the model's simulations (for the ACF, the dotted line almost coincides with the zero line). What do we learn?

We observe (a) that the RBC model ACF for output growth is flat at 0: there is no serial correlation of output growth generated by the model. In other words, model output growth independently distributed, (b) the spectrum is flat (like the one of a white noise), meaning that no periodic component is more important than another in explaining output growth. We do not observe a peak at business cycle frequencies.

We know that this model does a decent job in replicating some features of the business cycle (variance, comovements, autocorrelation of HP cycles components). When we consider the model's performances with respect to those particular dimesions of the data (ACF and spectrum of output growth), the model does a very poor job.

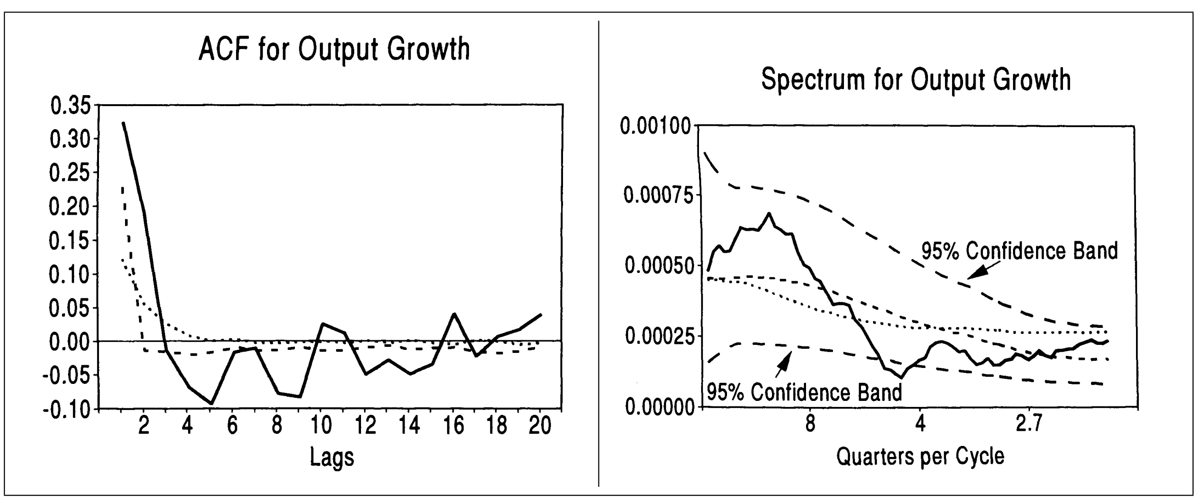
6 – The results displayed on Figure 2 are obtained by simulating a variant of the Christiano-Eichenbaum model that introduces adjustment cost to capital and labor. In that model, the production function is given by $\ln y_t = \ln[f(k_t, a_t n_t)] - (\alpha_k/2)[\Delta k_t/k_{t-1}]^2 - (\alpha_n/2)[\Delta n_t/n_{t-1}]^2$. How are the results changed? How do you explain those changes?
The model with adjustment costs to inputs (capital and labor) displays much more persistence (a) the ACF for output growth shows positive order 1 and 2 serial correlation and (b) the spectral density is not flat, but peaks at business cycle frequencies. Adjustment costs are adding internal persistence mechanism to the model. Because of adjustment costs, the response of the economy to shocks is more spread across time, therefore more persistent.

Figure 1 : Autocorrelation and Spectral Density (1)



Note: Solid lines show sample moments, and dotted lines show moments that are generated by the Christiano- Eichenbaum model

Figure 2 : Autocorrelation and Spectral Density (2)



Solid lines show sample moments, dotted lines show moments generated by a model with adjustment costs to capital and labor (forget about the dashed lines).

Extract 1 : The Christiano-Eichenbaum model

We begin by outlining the Christiano-Eichenbaum model. In this model, there is a representative agent whose preferences are given by

$$(1) \qquad E_t\left\{\sum_{j=0}^{\infty}\beta^j\left[\ln(c_{t+j})+\gamma(N-n_{t+j})\right]\right\}$$

where c_t is consumption, N is the total endowment of time, n_t is labor hours, and β is the subjective discount factor. Following Christiano and Eichenbaum, we assume that $\beta = 1.03^{-0.25}$ and $\gamma = 0.0037$.

There is also a representative firm that produces output by means of a Cobb-Douglas production function:

$$(2) \qquad y_t = k_t^{\theta}(a_t n_t)^{1-\theta}$$

where y_t is output, k_t is the capital stock, and a_t is a technology shock. The capital stock obeys the usual law of motion:

$$(3) \qquad k_{t+1} = (1-\delta)k_t + i_t$$

where δ is the depreciation rate and i_t is gross investment. Christiano and Eichenbaum (1992) estimate that $\theta = 0.344$ and $\delta = 0.021$.

The model is driven by technology and government spending shocks. Technology shocks are assumed to be difference-stationary, but none of our results depends on this assumption.[7] We initially assume that technology shocks follow a random walk with drift and that government spending shocks evolve as a persistent AR(1) process around the stochastic technology trend:

$$(4) \qquad (1-L)\ln(a_t) = \mu + \varepsilon_{at}$$

$$(5) \qquad \ln(g_t) - \ln(a_t) = \bar{g} + \varepsilon_{gt}/(1-\rho L)$$

where g_t denotes government spending and ε_{at} and ε_{gt} are the technology and government spending innovations, respectively. Later we consider models with more complicated impulse dynamics. Christiano and Eichenbaum (1992) estimate $\mu = 0.004$, $\bar{g} = 0.177$, and $\rho = 0.96$. We assume that the relative volatility of the two shocks is the same as in Christiano and Eichenbaum (1992), but we rescale the innovation variances so that the model matches the sample variance of per capita GNP growth. This yields $\sigma_a = 0.0097$ and $\sigma_g = 0.0113$.

The remainder of this section considers whether these models are consistent with the stylized facts discussed in the previous section. Our statistical approach is based on Monte Carlo simulation. The models were used to generate artificial data over a time horizon of 140 quarters, which matches the length of the sample period, 1954:1–1988:4. Each model was simulated 1,000 times. Autocorrelation and impulse-response functions were estimated for each artificial sample, and the results were collected into empirical probability distributions. The empirical distributions were then used to calculate the probability of observing the statistics estimated from U.S. data under the hypothesis that the data were generated by a particular RBC model.